1. $z = w^T \vec{x} + b = \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} + (-2) = (2-3) - 2 = -3$

2. $z = wx + b = (-0.5)(4) + (1) = -1$; $\hat{y} = \sigma(z) = \frac{1}{1+e^{-1}} = 0.73$

3. $z_1 = W_1 \vec{x} + \vec{b}_1 = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$; $h = \mathrm{ReLU}(z_1) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

$y = W_2 h + b_2 = \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix} + 0 = -3$

4. $L = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}(-1-2)^2 = 9/2$

5. $L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \frac{\partial L}{\partial \hat{y}} = \frac{1}{2} \cdot 2(\hat{y} - y) \cdot 1 = \hat{y} - y$

6. $\hat{y} = w^T \vec{x} + b$; $L = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}(w^T \vec{x} + b - y)^2$

$\begin{bmatrix} \partial L / \partial w_1 \\ \partial L / \partial w_2 \end{bmatrix} = \frac{1}{2} \cdot 2 (w^T \vec{x} + b - y)(\vec{x}) = \left(\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 1 - 0 \right) \left(\begin{bmatrix} 1 \\ -2 \end{bmatrix} \right) = (-1) \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$\frac{\partial L}{\partial b} = \frac{1}{2} \cdot 2 (w^T \vec{x} + b - y) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 1 = -1$

7. $w_{new} = w_{old} - \eta \frac{\partial L}{\partial w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 0.2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.6 \end{bmatrix}$

$b_{new} = b_{old} - \eta \frac{\partial L}{\partial b} = 1 - 0.2(-1) = 1.2$

8. $z = wx + b = (2)(-1) + (-1) = -3$; $a = \mathrm{ReLU}(z) = 0$; $L = \frac{1}{2}(a - y)^2 = \frac{1}{2}(\mathrm{ReLU}(wx+b) - y)^2$

$\frac{\partial L}{\partial w} = 2 \cdot \frac{1}{2}(\mathrm{ReLU}(wx+b) - y)\left(\frac{\partial}{\partial w}(\mathrm{ReLU}(wx+b))\right) = \begin{cases} (\mathrm{ReLU}(wx+b) - y)\,x, & x > 0 \\ 0, & x < 0 \end{cases} = 0$

$\frac{\partial L}{\partial b} = 2 \cdot \frac{1}{2}(\mathrm{ReLU}(wx+b) - y)\left(\frac{\partial}{\partial w}(\mathrm{ReLU}(wx+b))\right) = 0$

9. $\frac{\partial L}{\partial b_0} = \frac{\partial}{\partial b_0}\left(\frac{1}{2}(wx+b-y)^2\right) = wx+b-y = (0)(2) + (0) - (1) = -1$; $b_1 = b_0 - \eta\left(\frac{\partial L}{\partial b_0}\right) = 0 - (0.1)(-1) = 0.1$

$\frac{\partial L}{\partial w_0} = \frac{\partial}{\partial w_0}\left(\frac{1}{2}(wx+b-y)^2\right) = (wx+b-y)(x) = ((0)(2) + (0) - (1))(2) = -2$; $w_1 = w_0 - \eta\left(\frac{\partial L}{\partial w_0}\right) = (0) - (0.1)(-2) = 0.2$

$\frac{\partial L}{\partial b_1} = (0.2)(2) + (0.1) - (1) = -0.5$; $b_2 = b_1 - \eta\left(\frac{\partial L}{\partial b_1}\right) = 0.1 - (0.1)(-0.5) = 0.15$

$\frac{\partial L}{\partial w_1} = ((0.2)(2) + (0.1) - (1))(2) = -1$; $w_2 = w_1 - \eta\left(\frac{\partial L}{\partial w_1}\right) = 0.2 - (0.1)(-1) = 0.3$

10. $\mathrm{softmax}(s) = \mathrm{softmax}\left(\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}\right) = \left(\begin{bmatrix} \frac{1}{1+e+e^2} \\ \frac{e}{1+e+e^2} \\ \frac{e^2}{1+e+e^2} \end{bmatrix}\right)$

11. Forward pass uses some set biases, weights, and activation functions to calculate a predicted value from the model. Backprop is the error calculation, where the loss is used to calculate gradients for weight and biases updates.

12. Activation functions introduces non-linearity to linear functions. Without them, the values passed into the next layers would blow up and not be meaningful for learning.

13. If the learning rate is too large, the optimal weights and biases may be skipped. If too small, learning would take too long.

14. Vanishing gradients are when the gradients are computed as very small numbers, and the model will learn less effectively due to this. It happens when the subsequent inputs reach saturated regions in the activation functions.

15. ReLU is often preferred because it avoids the vanishing gradient problem that sigmoid may lead to.